

# Tal Ben Ami

## Performance Engineer | Parallel Computing & AI Workload Optimization

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Portfolio: [TalBenAmi.com](https://TalBenAmi.com) · LinkedIn: [linkedin.com/in/tal-ben-ami](https://linkedin.com/in/tal-ben-ami) · GitHub: [github.com/TalBenAmii](https://github.com/TalBenAmii)

Available 3–4 days/week; full-time upon graduation in March 2027.

Performance engineer at Intel focused on firmware latency, CPU bottlenecks, and reproducible benchmarking. Project and coursework experience includes CUDA kernel optimization with NVIDIA Nsight Systems/Compute, LLM inference benchmarking on Kubernetes, and memory-efficient Vision Transformer pre-training.

## EXPERIENCE

### Intel | Performance Engineer, TDX Design Team

Jan 2026 - Present

- Found and reduced affected SEAMCALL/TDCALL API latencies by ~33–71% on a newer server platform by resolving pipeline bottlenecks involving MOVDIR64B/non-temporal memory operations and SEAMCALL/SEAMRET transitions. Latencies moved from 20–70% above Granite Rapids to 20–50% below.
- Owned the team's performance infrastructure and tooling for measuring SEAMCALL and TDCALL API latency. Refactored benchmarks to reduce run-to-run variation and improve measurement reproducibility.
- Built test automation integrating an LLM via the OpenAI API to analyze Git diffs between releases and help identify causes of performance regressions.

## SELECTED PROJECTS

### Ilm-d: LLM Inference Benchmarking & Agent Automation

[Harness](#) · [MCP](#)

- Built an agent harness for Ilm-d, an open-source distributed inference project, translating workload goals into benchmark configurations and orchestrating Kubernetes Jobs and parameter sweeps. Exposed workflows through an MCP server.
- Implemented schema-validated analysis of time to first token (TTFT), inter-token latency (ITL), and throughput, with SLO checks and Pareto comparisons.

### CoLA + GaLore: Memory-Efficient Vision Transformer Training

[GitHub](#)

- Developed a PyTorch implementation of CoLA low-rank layers and GaLore gradient projection from research papers. Profiling a large ViT model showed ~70% lower peak allocated GPU memory (>20 to 6.45 GiB) with ~25% higher training time per iteration versus the AdamW baseline.
- On CIFAR-10, a smaller ViT model used ~80% less memory with validation accuracy ~2 percentage points below baseline.

### Computer Architecture Simulation Suite

[GitHub](#)

- Built L1/L2 cache and two-level branch-predictor simulators (LRU, write-back/write-allocate, L-share/G-share), plus an out-of-order RAW dependency analyzer. Evaluated miss rates, access times, and latency-aware critical paths.

### DOMINO | 2nd Place | ILS 4,000 Team Prize

[GitHub](#)

- Developed a Flutter emergency-response application with Firebase as its database backend, Hebrew speech transcription, LLM parsing of casualty reports, synced field/command/hospital dashboards, and medical handoff PDFs.

## TECHNICAL SKILLS

**Languages & systems:** C/C++, Python, x86 assembly, Bash, Linux, computer architecture, memory hierarchy, firmware analysis.

**Performance & parallel computing:** Profiling, benchmarking, bottleneck and root-cause analysis, latency/throughput analysis, parallel programming, CUDA kernel optimization, NVIDIA Nsight Systems/Compute, OpenMP, OpenCL.

**AI & modeling:** PyTorch, Hugging Face Transformers, NumPy, LLM inference benchmarking, memory-efficient training, computer architecture simulation.

**Networking & distributed computing:** Academic background in RDMA/RoCE, InfiniBand, and MPI.

**Tools & infrastructure:** Docker, Kubernetes, Git, GitHub Actions (CI/CD), Make, CMake.

**Deep learning optimization:** Familiar with vLLM, KV/prefix caching, PagedAttention, continuous batching, chunked prefill, speculative decoding, and data/tensor/pipeline parallelism.

## EDUCATION & HONORS

### Technion, Israel Institute of Technology

2022 - Mar 2027 (expected)

B.Sc. in Computer Engineering | Average: 86.6/100 · **Dean's List, Spring 2023/24**

### Selected coursework:

High Performance Parallel Computing (99) · Deep Learning on Computational Accelerators (94) · Computational Accelerators and Accelerated Systems (93) · Performance Engineering of Software Systems (91) · Concurrent and Distributed Programming for Data Processing and Machine Learning (90).



## Semester Excellence

THIS IS TO CERTIFY THAT

Tal Ben Ami

ID: 212525257

Mr. Tal Ben Ami was included in the Dean's list of honors for scholastic achievement in the Spring semester of 2023/24.

Dr. Efrat Nativ-Ronen



Academic Secretary Undergraduate Studies

Haifa, 25.09.2026



Transcript of **TAL BEN AMI** ID: **212525257**

who studies

for the degree **BACHELOR OF SCIENCE IN COMPUTER ENGINEERING**

in the Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering

and accumulated **142.5** credit points out of **158.5** credit points required for a degree, with a GPA of **86.6**

| SUBJECT                                          | CREDITS | GRADE                    | SEMESTER         |
|--------------------------------------------------|---------|--------------------------|------------------|
| 00440102 Safety in Ee Labs.                      | 0       | Pass                     | 2022-2023 Winter |
| 00440105 Theory of Electronic Circuits           | 4       | 90                       | 2023-2024 Winter |
| 00440127 Basics of Semiconductor Devices         | 3.5     | 96                       | 2023-2024 Spring |
| 00440131 Signals and Systems                     | 5       | 83                       | 2023-2024 Spring |
| 00440157 Electrical Engineering Lab 1a           | 2       | 96                       | 2024-2025 Summer |
| 00440167 Lab. Project a                          | 4       | 97                       | 2025-2026 Winter |
| 00440252 Digital Systems and Computer Structure  | 5       | 80                       | 2022-2023 Spring |
| 00450002 An Overview of Electrical and Comp.Eng. | 1       | 99                       | 2022-2023 Winter |
| 00460267 Computer Architecture                   | 3       | 90                       | 2024-2025 Winter |
| 00460278 Computational Accelerators and Accelera | 3       | 93                       | 2025-2026 Spring |
| 00460279 High Performance Parallel Computing     | 3       | 99                       | 2025-2026 Spring |
| 01030015 Introductory Mathematics                | 0       | Exemption without points | 2022-2023 Winter |
| 01040013 Differential and Integral Calculus 2T   | 5.5     | 77                       | 2022-2023 Spring |
| 01040016 Algebra 1/Extended                      | 5       | 85                       | 2022-2023 Winter |
| 01040034 Introduction to Probability H           | 3.5     | 81                       | 2022-2023 Summer |
| 01040036 DIFFERENTIAL AND INTEGRAL CALCULUS 1T   | 5       | 63                       | 2022-2023 Winter |
| 01040038 ALGEBRA 2M                              | 2.5     | Pass                     | 2022-2023 Spring |
| 01040134 Modern Algebra H                        | 2.5     | 79                       | 2024-2025 Winter |
| 01040136 ORDINARY DIFFERENTIAL EQUATIONS M       | 4       | Pass                     | 2022-2023 Spring |
| 01040214 Fourier Series and Integral Transforms  | 2.5     | 84                       | 2023-2024 Winter |
| 01040215 Complex Functions a                     | 2.5     | 78                       | 2023-2024 Winter |
| 01040220 Partial Differential Equations/T        | 2.5     | 90                       | 2023-2024 Winter |
| 01130013 Introductory Physics 1                  | 0       | Exemption without points | 2022-2023 Winter |

(E): The course was taught in English

Minimal Passing Grade is 55

Grade Scale: 0 – 100

Haifa, 25.09.2026

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|------------------------------------------------------------------------------------------|---------|--------------------------|------------------|
| 01130014 Introductory Physics 2                                                          | 0       | Exemption without points | 2022-2023 Winter |
| 01140071 Physics 1M                                                                      | 3.5     | 88                       | 2022-2023 Winter |
| 01140073 Quantum physics for engineering                                                 | 3.5     | 68                       | 2023-2024 Spring |
| 01140075 Physics 2Mm                                                                     | 5       | 86                       | 2022-2023 Spring |
| 01230015 Introductory Chemistry                                                          | 0       | Exemption without points | 2023-2024 Winter |
| 02340117 Introduction to Computer Science H                                              | 4       | 88                       | 2022-2023 Winter |
| 02340118 Computer Organization and Programming                                           | 3       | 100                      | 2023-2024 Spring |
| 02340123 Operating Systems                                                               | 4.5     | 93                       | 2024-2025 Winter |
| 02340124 Introduction to Systems Programming                                             | 4       | 96                       | 2023-2024 Winter |
| 02340141 Combinatorics for Cs                                                            | 3       | 65                       | 2023-2024 Winter |
| 02340218 Data Structures 1                                                               | 3       | 89                       | 2023-2024 Spring |
| 02340247 Algorithms 1                                                                    | 3       | 87                       | 2024-2025 Winter |
| 02340901 Workshop in Competitive Programming                                             | 3       | 90                       | 2024-2025 Spring |
| 02360013 SELECTED TOPICS IN PERFORMANCE ENGINEERING OF SOFTWARE SYSTEMS                  | 2       | 91                       | 2025-2026 Spring |
| 02360334 Introduction to Computer Networks                                               | 3       | 94                       | 2024-2025 Spring |
| 02360343 Theory of Computation                                                           | 3       | 80                       | 2024-2025 Spring |
| 02360370 Concurrent and Distributed Programming for Data Processing and Machine Learning | 3       | 90                       | 2025-2026 Winter |
| 02360496 Reverse Engineering                                                             | 3       | 94                       | 2024-2025 Spring |
| 02360501 Introduction to Artificial Intelligence                                         | 3       | 92                       | 2024-2025 Spring |
| 02360700 Software Design                                                                 | 3       | 73                       | 2025-2026 Winter |
| 02360781 DEEP LEARNING ON COMPUTATION (E)                                                | 3       | 94                       | 2025-2026 Winter |
| 03200101 Community Program 1                                                             | 1       | Exemption with points    | 2022-2023 Winter |
| 03200102 Community Program 2                                                             | 1       | Exemption with points    | 2022-2023 Winter |
| 03240032 Advanced English - a                                                            | 0       | Exemption without points | 2023-2024 Winter |
| 03240033 Technical English-Advanced B                                                    | 3       | Exemption with points    | 2022-2023 Winter |
| 03240053 Hebrew 3                                                                        | 0       | Exemption without points | 2023-2024 Winter |

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| SUBJECT                                             | CREDITS | GRADE | SEMESTER         |
|-----------------------------------------------------|---------|-------|------------------|
| 03240540 Legal Aspects of Busine.Entrepreneurship   | 2       | 96    | 2025-2026 Spring |
| 03240879 Issues in Contemporary Israeli Society (E) | 2       | 97    | 2025-2026 Winter |
| 03250000 SEXUAL HARASSMENT PREVENTION               | 0       | Pass  | 2023-2024 Winter |
| 03940805 Physical Education - Athletics             | 1       | 98    | 2025-2026 Winter |
| 03940808 Physical Education-for New Students        | 1       | 95    | 2022-2023 Winter |

END OF TRANSCRIPT

I hereby confirm that the information above is correct

Dr. Efrat Nativ Ronen,  
Academic Secretary, Undergraduate Studies

